Full dataset of several mechanical tests on an S355 steel sheet as reference data for digital representations

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Abstract

The dataset provided in this repository comprises data obtained from a series of characterization tests performed to a sheet of typical S355 (material number: 1.0577) structural steel (designation of steel according to DIN EN 10025-2:2019 [1]). The tests include methods for the determination of mechanical properties such as, e.g., tensile test, Charpy test and sonic resonance test. This dataset is intended to be extended by the inclusion of data obtained from further test methods. Therefore, the entire dataset (concept DOI) comprises several parts (versions), each of which is addressed by a unique version DOI.

The data were generated in the frame of the digitization project Innovationplatform Material*Digital* (PMD) [2] which, amongst other activities, aims to store data in a semantically and machine understandable way. Therefore, data structuring and data formats are focused in addition to aspects in the field of material science and engineering (MSE). Hence, this data is supposed to provide reference data as basis for experimental data inclusion, conversion and structuring (data management and processing) that leads to semantical expressivity as well as for MSE experts being generally interested in the material properties and knowledge.

1. Content

In this section, the files given in the repository are listed and described. There are data (files) on the general consideration of the material (steel sheet) and test planning as well as specific data with respect to the associated test methods.

The folder structure the files are given in is selected deliberately such that a typical structure of measurement data is represented, including a preferable differentiation of primary data, secondary data and metadata. Such a structure is intended to be introduced as best practice for the publication of experimental data (measurement data) obtained from characterization methods.

Type / Description	**Filename**
General description of data	_steel_sheet_test_data_mission_abstract_content.pdf
Overview of extraction of test pieces	test_piece_extraction_sketch.pdf
Declaration of performance of the material used (provided by a manufacturer) in English, German, French, for more information, see section 2 - Remarks	Declaration_of_performance_S355_1.0577.pdf

Tensile test

All tensile tests including analysis were performed in accordance with DIN EN ISO 6892-1:2020 [3] with the friendly assistance of BAM colleagues Kaspar Gorges, Nico A. Staudler, Michelle Schumann and Jennifer Jacob.

Type / Description	Filename
Primary data – Tensile test measurement data in comma separated value (CSV) format	01_primary_data\Zd2.csv 01_primary_data\Zd3.csv 01_primary_data\Zx1.csv 01_primary_data\Zx2.csv 01_primary_data\Zx3.csv 01_primary_data\Zx4.csv 01_primary_data\Zy1.csv 01_primary_data\Zy2.csv 01_primary_data\Zy3.csv 01_primary_data\Zy4.csv
Secondary data – Diagrams created from plotting the relevant primary data (overview: '[test_piece_name]_Sp-Weg.pdf'; detailed look for yield strength determination: '[test_piece_name]-ReH.pdf')	02_secondary_data\diagrams\Zd2_Sp-Weg.pdf 02_secondary_data\diagrams\Zd2-ReH.pdf 02_secondary_data\diagrams\Zd3_Sp-Weg.pdf 02_secondary_data\diagrams\Zd3-ReH.pdf 02_secondary_data\diagrams\Zx1_Sp-Weg.pdf 02_secondary_data\diagrams\Zx1-ReH.pdf 02_secondary_data\diagrams\Zx2_Sp-Weg.pdf 02_secondary_data\diagrams\Zx2-ReH.pdf 02_secondary_data\diagrams\Zx3_Sp-Weg.pdf 02_secondary_data\diagrams\Zx3-ReH.pdf 02_secondary_data\diagrams\Zx4_Sp-Weg.pdf 02_secondary_data\diagrams\Zx4-ReH.pdf 02_secondary_data\diagrams\Zy1_Sp-Weg.pdf 02_secondary_data\diagrams\Zy1-ReH.pdf 02_secondary_data\diagrams\Zy2_Sp-Weg.pdf 02_secondary_data\diagrams\Zy2-ReH.pdf 02_secondary_data\diagrams\Zy3_Sp-Weg.pdf 02_secondary_data\diagrams\Zy3-ReH.pdf 02_secondary_data\diagrams\Zy4_Sp-Weg.pdf 02_secondary_data\diagrams\Zy4-ReH.pdf
Secondary data – Test protocols (German language) and analysis results of all performed tensile tests in XLSX data format	02_secondary_data\results_protocols\ Protokoll-Zugversuch RT Zd.xlsx 02_secondary_data\results_protocols\ Protokoll-Zugversuch RT Zx.xlsx 02_secondary_data\results_protocols\ Protokoll-Zugversuch RT Zy.xlsx 02_secondary_data\results_protocols\tensile_test_results.xlsx
Metadata – Additional information on the equipment used during the tests; list and description of testing and measuring instruments	03_metadata\Testing_measuring_instruments_tensile_test.pdf

Charpy test

All Charpy tests were performed in accordance with DIN EN ISO 148-1:2017 [4] with the friendly assistance of BAM colleagues Wolfram Baer, Peter Wossidlo, Michelle Schumann and Jennifer Jacob.

Type / Description	**Filename**
Primary data – Charpy test measurement data in XLSX format as protocol including measurement values and partly metadata information	01_primary_data\Charpy_protocol_KX_test_pieces.xlsx 01_primary_data\Charpy_protocol_KY_test_pieces.xlsx
Secondary data – Test results of Charpy tests	02_secondary_data\Charpy_test_results.xlsx
Metadata – Additional information on the equipment used during the tests; list and description of testing and measuring instruments	03_metadata\Testing_measuring_instruments_charpy_test.pdf

Sonic resonance test

All sonic resonance tests were performed in accordance with ASTM E 1875:2020 [5] including the consideration of thermal expansion with the friendly assistance of BAM colleague Faruk Bayram.

Type / Description	**Filename**
Primary data – sonic resonance test measurement data in XLSX format including test piece dimension measurements	01_primary_data\5.22_430_ERX1_ASTM_E1875.xlsx 01_primary_data\5.22_430_ERX2_ASTM_E1875.xlsx 01_primary_data\5.22_430_ERY1_ASTM_E1875.xlsx 01_primary_data\5.22_430_ERY2_ASTM_E1875.xlsx
Secondary data – Test and analysis results (German language) of all performed sonic resonance tests in XLSX data format	02_secondary_data\results_sonic_resonance_tests.xlsx
Metadata – Additional information on the equipment used during the tests; list and description of testing and measuring instruments	03_metadata\Testing_measuring_instruments_sonic_resonance_test.pdf

2. Remarks

Data structure:

If files contained information that could be ascribed to several data categories (primary, secondary, meta), the file was only given once in one of the according categories to avoid redundant data storage.

Information content:

The declaration of performance document including data on the chemical composition given in the dataset is that of a comparable material (batch) of a S355J2 (1.0577) steel but not necessarily referring to the lot the tested steel sheet was produced from. However, this steel is usually categorized regarding its mechanical properties as given in the declaration of performance document.

Tensile Test:

As typical for clip-on extensometers exhibiting a limited measuring range and in accordance with the testing standard [3], the extensometer used for extension measurement was detached from test pieces during the tensile test after conveniently having reached the region of plastic deformation (especially after the determination of, e.g., the slope in the elastic region of the stress-strain curve, yield strength and proof strength values). In the further course of the test, the crosshead separation was used as elongation data. Therefore, the number of values of percentage extension is lower than the overall number of measurement values for time, load and crosshead separation in primary data and a kink can be noticed in the stress-strain curves (secondary data) for every test.

3. Acknowledgments

For the support in test execution and analysis, the assistance of Kaspar Gorges, Nico A. Staudler (tensile tests), Michelle Schumann, Jennifer Jacob (tensile and Charpy tests), Wolfram Baer, Peter Wossidlo (Charpy test) and Faruk Bayram (sonic resonance tests) is gratefully acknowledged. The authors particularly appreciate valuable discussions with and the help of Prof. Pedro D. Portella in test planning and data assessment. Valuable discussions on the structuring of data involving participants of the Platform Material*Digital* (PMD) as well as of the ad-hoc working group on digitization of the DIN standardization committee “NA 062-01-42 AA” are especially acknowledged.

Funding:
The authors M.S. and B.S. gratefully acknowledge the financial support of the Federal Ministry of Education and Research (BMBF) through the project ‘Innovation platform MaterialDigital (PMD)’ [grant number 13XP5094E].

References

[1] DIN EN 10025-2, Hot rolled products of structural steels - Part 2: Technical delivery conditions for non-alloy structural steels; German version EN 10025-2:2019, Deutsches Institut für Normung (DIN), October 2019.

[2] materialdigital.de

[3] DIN EN ISO 6892-1:2020-06, Metallic materials - Tensile testing - Part 1: Method of test at room temperature; German version EN ISO 6892-1:2020-06, Deutsches Institut für Normung (DIN), June 2020.

[4] DIN EN ISO 148-1, Metallic materials - Charpy pendulum impact test - Part 1: Test method (ISO 148-1:2016); German version EN ISO 148-1:2016, Deutsches Institut für Normung (DIN), May 2017.

[5] ASTM E 1875a, Standard Test Method for Dynamic Young's Modulus, Shear Modulus, and Poisson's Ratio by Sonic Resonance, American Society for Testing and Materials International (ASTM), 2020.